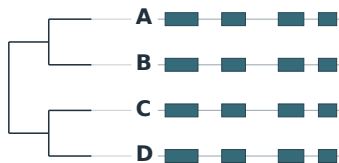


From genomic loci to local structural evolution

Evidence, states and evolutionary summaries at the same homologous interval

A Orthologous loci



Genome + annotation
Rooted species tree supplied

B Sequence correspondence

Genomic locus

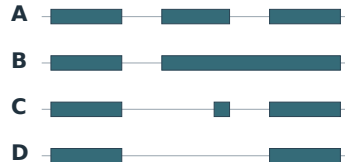


Coding alignment → genome



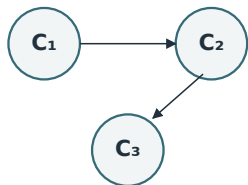
Copy · order · strand · flanks

C Local configurations



Observed · partial · unknown
Coexisting configurations

D Admissible state graph



Observed + admissible states

E Tree-based inference

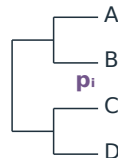


Parsimony
min-cost paths

CTMC
sum over states

Species tree supplied
Local structure reconstructed

F Ancestral reconstruction



Marginal state probabilities
Expected events along branches